BATTERY MONITORING SYSTEM

Battery monitoring systems.

YELLOW solution catalogue / EN



Each battery. Every string. All locations.

Standby power monitoring for data centres, telecoms, substations and industrial sites.

SELECTING A SYSTEM

From application to system.

YELLOW model	Application	Batteries / strings
PBMS2000	Telecom sites	Up to 120 batteries 1 × 120 or 2 × 60 batteries
PBAT-Gate	UPS and small data centres	Up to 480 batteries Up to 4 strings of 120 batteries
PBMS9000	Large data centres	Up to 640 batteries · 8C version Up to 8 strings of 80 batteries
PBMS9000Pro	Substations and industry	Up to 420 batteries Up to 6 strings of 70 batteries

- **Individual battery monitoring**

  Voltage, temperature and impedance, depending on the selected sensors.

- **String monitoring**

  Charge/discharge current, string voltage and additional measurements as specified.

- **Supervision**

  Local display, history, events and data transfer to site supervisory systems.

SOLUTION / TELECOM SITES

YELLOW PBMS2000

Telecom battery monitoring with automatic balancing.



Batteries	VRLA · 24 / 48 V systems
Capacity	Up to 120 batteries
Strings	1 × 120 or 2 × 60 batteries
Integration	Modbus RTU / TCP, SNMP

Maximum battery counts apply to the specified version. Sensors, power supplies and interfaces are selected for the project.

COMPONENTS AND SPECIFICATIONS

YELLOW PBMS2000

Battery sensors	YELLOW PBAT61
String measurement	Hall sensor for string current measurement
Display	HMI / YELLOW Monitor

Technical parameters

Power supply	24 / 48 V DC; 20-60 V DC
Communication	Ethernet / RS485
Controller consumption	< 2 W

Configuration requirements

Confirm battery type and voltage, batteries per string, parallel string count, power supply, equipment layout and supervisory protocols.

Voltage, current and environmental sensors, cables, HMI and alert channels are specified separately. SOC/SOH estimates do not replace capacity testing.

SOLUTION / UPS AND SMALL DATA CENTRES

YELLOW PBAT-Gate

Web interface, 12-month history and data acquisition via PBAT600Pro.



Batteries	Lead-acid batteries
Capacity	Up to 480 batteries
Strings	Up to 4 strings of 120 batteries
Integration	Modbus TCP, SNMP

Maximum battery counts apply to the specified version. Sensors, power supplies and interfaces are selected for the project.

COMPONENTS AND SPECIFICATIONS

YELLOW PBAT-Gate

Battery sensors	YELLOW PBAT51
String measurement	PBAT600Pro and Hall sensor per battery string
Display	HMI / YELLOW Monitor

Technical parameters

Controller power supply	18-36 V DC
Data history	12 months
String acquisition	PBAT600Pro

Configuration requirements

Confirm battery type and voltage, batteries per string, parallel string count, power supply, equipment layout and supervisory protocols.

Voltage, current and environmental sensors, cables, HMI and alert channels are specified separately. SOC/SOH estimates do not replace capacity testing.

SOLUTION / LARGE DATA CENTRES

YELLOW PBMS9000

19″ / 1U controller with ring communication and dual power inputs.



Batteries	Lead-acid, including multi-pole batteries
Capacity	Up to 640 batteries · 8C version
Strings	Up to 8 strings of 80 batteries
Integration	Modbus RTU / TCP, SNMP, MQTT, IEC 61850

Maximum battery counts apply to the specified version. Sensors, power supplies and interfaces are selected for the project.

COMPONENTS AND SPECIFICATIONS

YELLOW PBMS9000

Battery sensors	YELLOW PBAT61
String measurement	PBMS-CM2 and Hall sensor per battery string
Display	HMI / YELLOW Monitor

Technical parameters

Mounting	19 inch / 1U
Dimensions W x D x H	483 x 206 x 44.5 mm
Controller consumption	< 15 W
Digital channels	6 DI / 6 DO

Configuration requirements

Confirm battery type and voltage, batteries per string, parallel string count, power supply, equipment layout and supervisory protocols.

Voltage, current and environmental sensors, cables, HMI and alert channels are specified separately. SOC/SOH estimates do not replace capacity testing.

SOLUTION / SUBSTATIONS AND INDUSTRY

YELLOW PBMS9000Pro

Bus-powered sensors. Leakage and electrolyte level monitoring are optional.



Batteries	VRLA / Ni-Cd, including multi-pole batteries
Capacity	Up to 420 batteries
Strings	Up to 6 strings of 70 batteries
Integration	Modbus RTU / TCP, SNMP, MQTT, IEC 61850

Maximum battery counts apply to the specified version. Sensors, power supplies and interfaces are selected for the project.

COMPONENTS AND SPECIFICATIONS

YELLOW PBMS9000Pro

Battery sensors	YELLOW PBAT71
String measurement	String current module and Hall sensor as specified
Display	HMI / YELLOW Monitor

Technical parameters

Mounting	19 inch / 1U
Dimensions W x D x H	483 x 206 x 44.5 mm
Power variants	V1: 48 V DC / V2: 220 V AC / V3: 110 V DC
Controller consumption	< 15 W

Configuration requirements

Confirm battery type and voltage, batteries per string, parallel string count, power supply, equipment layout and supervisory protocols.

Voltage, current and environmental sensors, cables, HMI and alert channels are specified separately. SOC/SOH estimates do not replace capacity testing.

SENSORS AND MEASUREMENTS

Individual batteries. One overview.

Series	Application	Measurements
YELLOW PBAT51	PBAT-Gate	U / T / R
YELLOW PBAT61	PBMS2000 / PBMS9000	U / T / R
YELLOW PBAT71	PBMS9000Pro	U / T / R

Sensor version

Select by chemistry and rated voltage. PBAT61 versions cover 2, 6 and 12 V batteries. PBAT71 covers 2 V lead-acid / 1.2 V Ni-Cd and 12 V lead-acid batteries.

String sensors

Hall sensors and measurement modules are selected for string current and wiring layout.

Optional measurements

Ambient temperature/humidity, insulation resistance, leakage and electrolyte level require the corresponding equipment.

MONITORING AND HMI

YELLOW Monitor

One interface across multiple locations.

- **Centralised monitoring**

  Up to 100,000 batteries in the platform configuration; project structure and server resources are specified during design.

- **History and events**

  Trends, alarm logs, centralised measurement storage and CSV export.

- **Alerts**

  SMS and email when the required channels and hardware/software configuration are available.

- **Local display**

  Optional 7-inch and 10.1-inch HMI panels. Compatibility depends on the selected controller.

The supervisory platform and local panel are separate system components.

PREPARATION AND COMMISSIONING

Commissioning workflow.

- **01 / Equipment and compatibility**

  Check controller, battery-compatible sensors, string modules, power supplies, cables and interfaces.

- **02 / Site wiring design**

  Qualified personnel must follow the approved wiring design for the selected equipment version.

- **03 / Addressing and battery map**

  Map sensor identifiers to physical batteries and strings.

- **04 / Measurement verification**

  Verify polarity and readings according to the equipment documentation. Record baseline values.

- **05 / Thresholds and supervision**

  Agree thresholds for the installed batteries and verify event delivery to the site supervisory system.

- **06 / Operational handover**

  Save the sensor map, settings, configuration backup and responsible contacts.